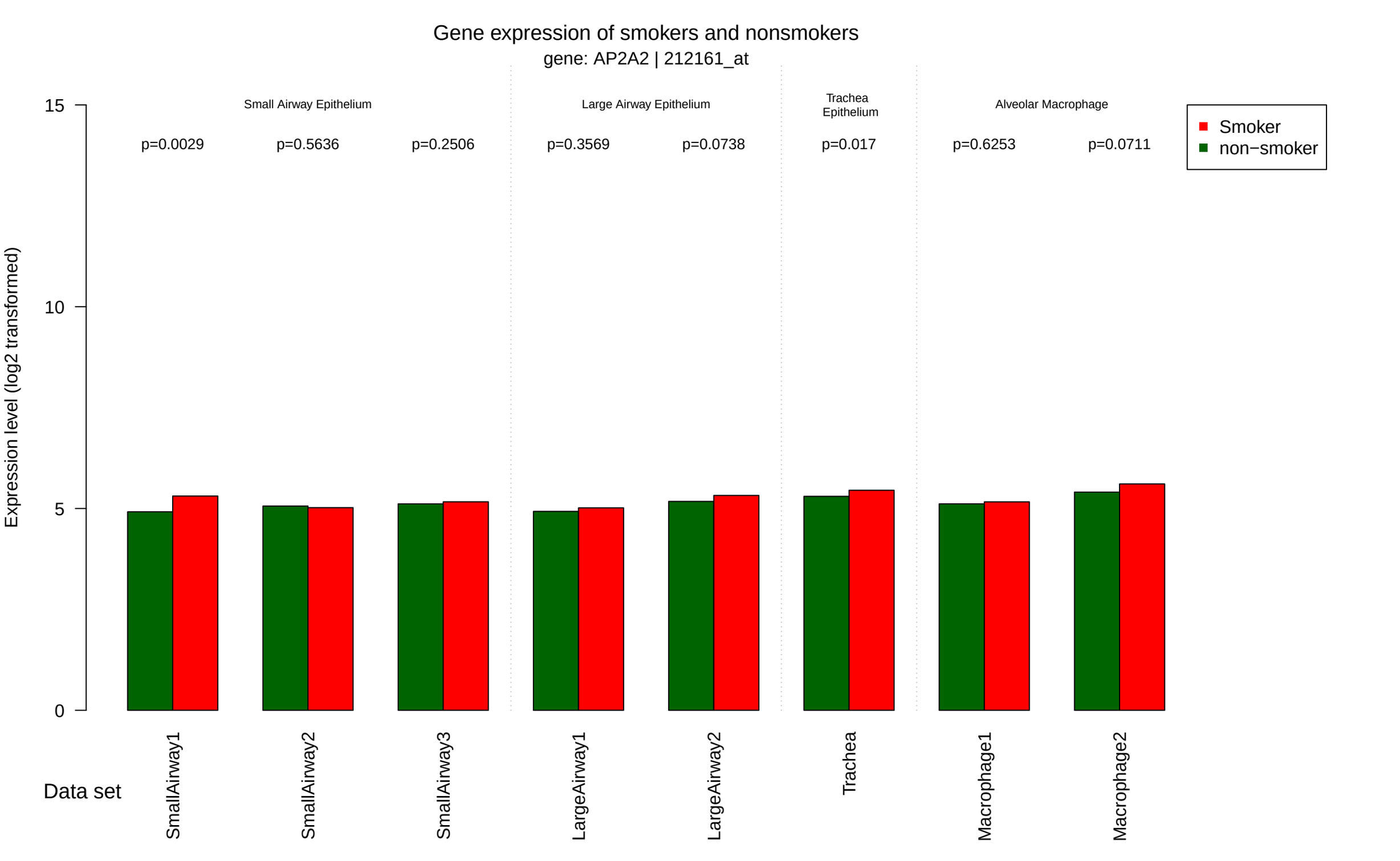
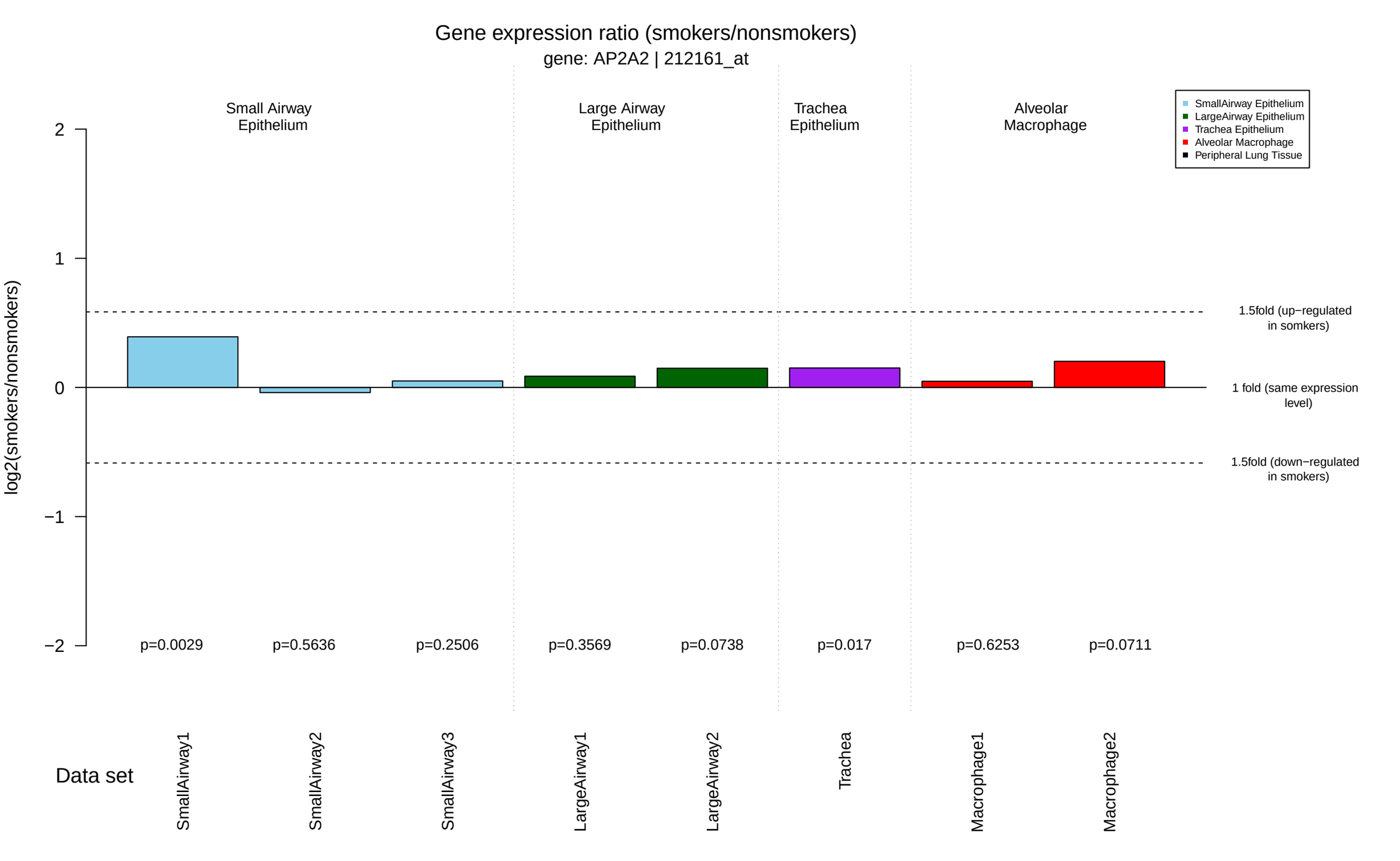
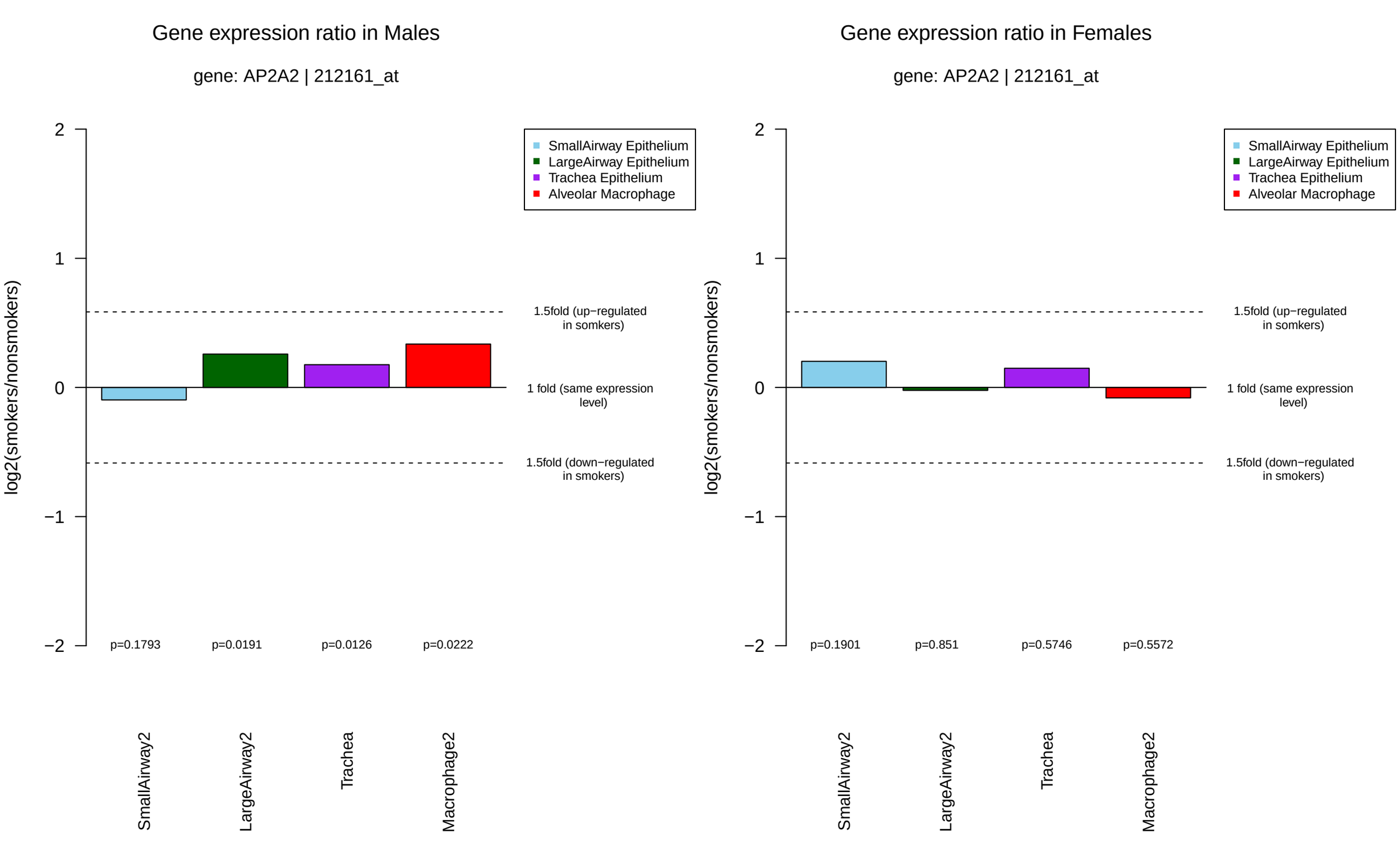
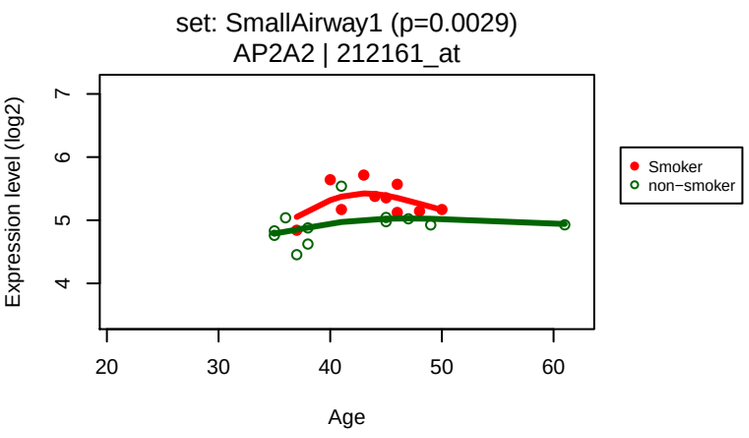
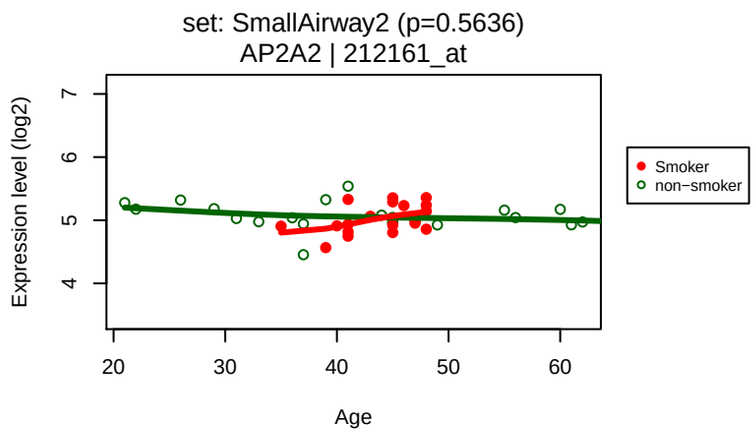
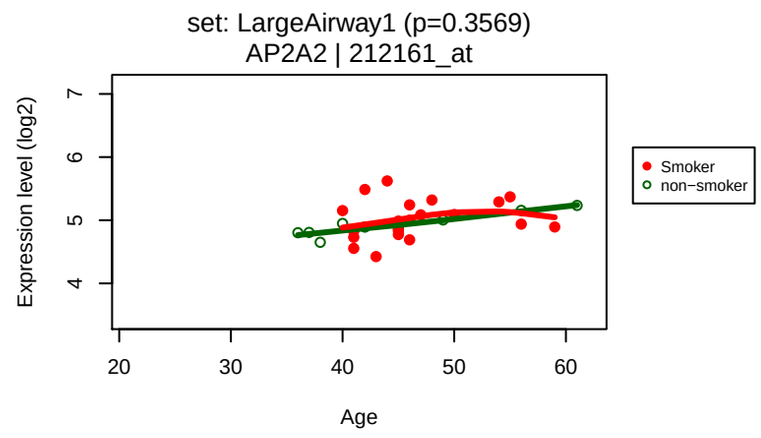

Gene expression of smokers and nonsmokers
gene: AP2A2 | 212161_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0029
p=0.5636
p=0.2506
p=0.3569
p=0.0738
p=0.017
p=0.6253
p=0.0711
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

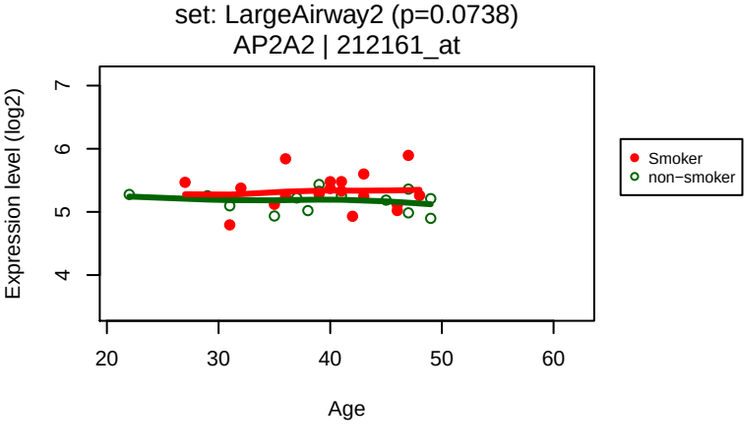

Gene expression ratio (smokers/nonsmokers)
gene: AP2A2 | 212161_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0029
p=0.5636
p=0.2506
p=0.3569
p=0.0738
p=0.017
p=0.6253
p=0.0711
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

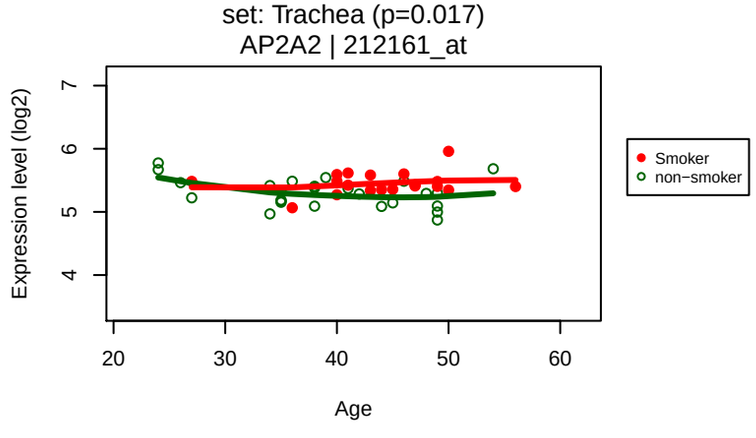

Gene expression ratio in Males
gene: AP2A2 | 212161_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1793
p=0.0191
p=0.0126
p=0.0222
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP2A2 | 212161_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1901
p=0.851
p=0.5746
p=0.5572
SmallAirway2
LargeAirway2
Trachea
Macrophage2

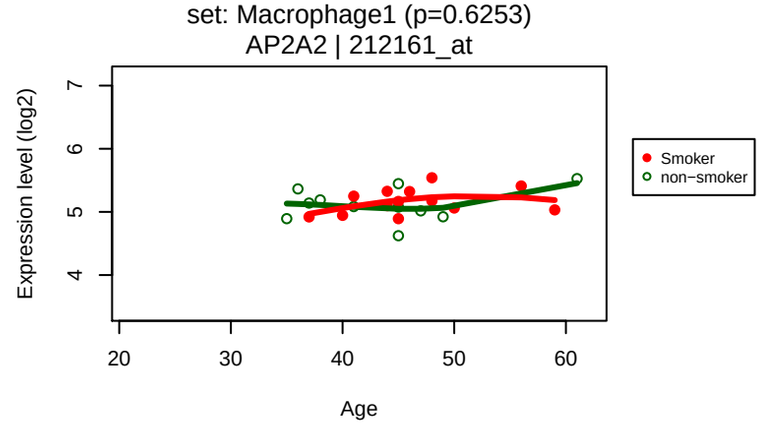

set: SmallAirway1 (p=0.0029)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker

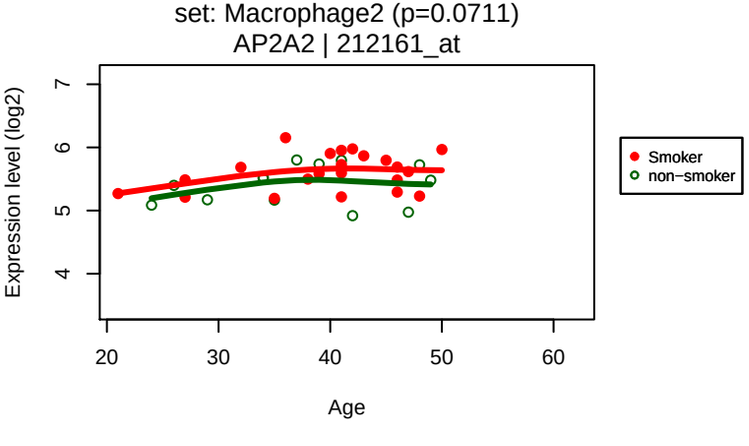

set: SmallAirway2 (p=0.5636)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker

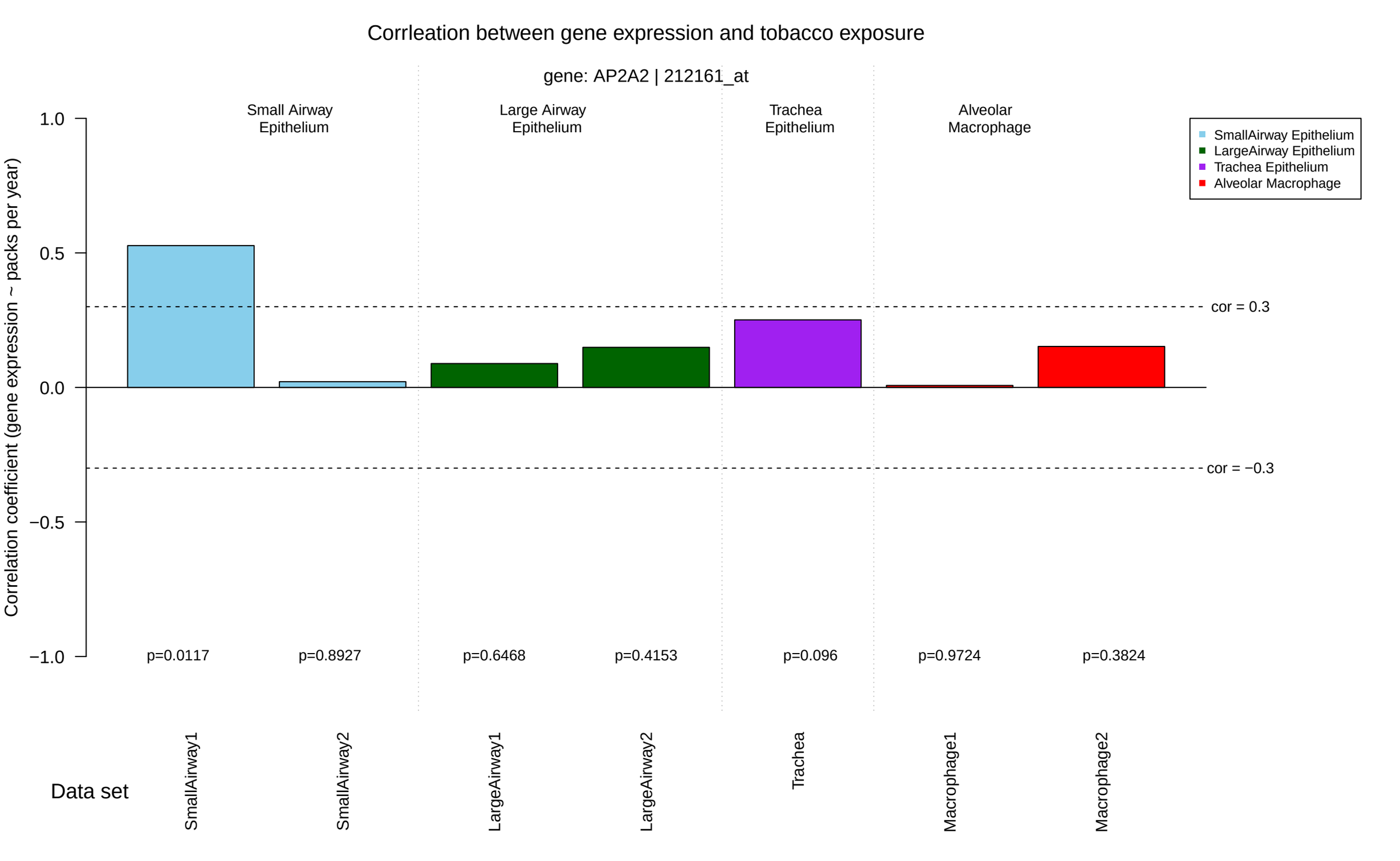

set: LargeAirway1 (p=0.3569)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0738)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.017)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6253)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0711)
AP2A2 | 212161_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP2A2 | 212161_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0117
p=0.8927
p=0.6468
p=0.4153
p=0.096
p=0.9724
p=0.3824
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2